

Shahjalal University of Science & Technology
Department of Computer Science & Engineering
2nd Year 1st Semester Lab Final Examination (Session 2021-2022)
Course: **CSE 234** Course Title: **Object Oriented Programming Language Lab**
Credits : **2.0** Full Marks : **60** Time : **1 Hour 20 Min**

- Instructions:**
- 1. If you have a **phone**, **USB flash drive**, or any digital **device** with you, keep them in your bag. If you are found with any of these items later you will be **expelled**.
 - 2. Create a **folder** on the **desktop** named after your **registration number** and put all the code files in it. If your answer code files are **not found** on the **desktop** then your answer codes will **not be taken**.
 - 3. At the start of each source **code file** write your **name**, and **registration number** in the **comment**.
 - 4. If you need to create package folders those also should be created inside this folder.
 - 5. All class, method, and variable **names** that are provided should be **kept the same** (case-sensitive) as it is.
 - 6. Keep your code **font size small** enough in VSCode at all times so that they are not readable to nearby students.
 - 7. Any type of **communication** (verbal or digital) with peers is strictly **prohibited**. If you try to see/show each other's code or try to talk, you will be **expelled**.
 - 8. Whenever exam time is finished, you should close all opened files/applications and **turn off** your **monitor**. **Do not shut down your computer**.

Q#1 **Marks: 20**

The following file **AQ1.java** is provided to you. You cannot change any code in this file.

```
1 // AQ1.java
2 abstract class A { int i; A(int i) {this.i = i;}
3     int m(int j) { return i+j; } abstract int m2(int j, int k); void m4(){}
4 class B extends A { B(int i) { super(i);}
5     int m2(int x, int y) { return x+y; }
6 }
7 public class AQ1 {
8     public static void main(String[] args) {
9         C c = new C(10, 20); B b = new B(5); System.out.println("Main Start");
10        c.m3("Java"); System.out.println("1:"+c.m2(6,6));
11        System.out.println("2:"+b.m2(6,6));
12        A a = b; System.out.println("3:"+a.m(15)); System.out.println("4:"+a.i);
13        a = c; System.out.println("5:"+a.m(15)); System.out.println("6:"+a.i);
14        a.m4(); c.i+=10; a.m4(); a.i+=10; a.m4(); new C(100,200);
15    }
16 }
```

Now write a Java Class named “C” in the **C.java** file, so that if we run the **main** method in **AQ1** it gives the following output:

```
Exam Started
Data Loaded
Main Start
Hello Java
1:72
2:12
3:20
4:5
5:25
6:10
20
30
40
Data Loaded
```

Consider the following class **AQ2** and the **main** method inside it. The following codes are provided to you in the **AQ2.java** file. You cannot change any code in this file.

```
1 // AQ2.java
2 package pkA;
3
4 class S implements pkA.p3.PC.IPC2 {}
5
6 public class AQ2 {
7     public static void main(String[] args) { System.out.println("Start");
8         pkA.p1.PA pa = new pkA.p1.PA(); System.out.println(pa.m("ABC"));
9         System.out.println(pa.m4(5)); System.out.println(pa.m4(10));
10        pkA.p3.PC.IPC ipc = pa; System.out.println(ipc.m4(20));
11        System.out.println(pkA.p1.p2.PB.m2("Java"));
12        pkA.p1.p2.PB pb = new pkA.p1.p2.PB(); System.out.println(pb.m(4,2));
13        System.out.println(pb.m(9,3));
14        pkA.p3.PC pc = pb; System.out.println(pc.m(7,1));
15        System.out.println(pkA.p3.PC.m3("XYZ"));
16        S s = new S(); System.out.println(s.m2(5));
17        pkA.p3.PC.IPC2 ipc2 = s; System.out.println(ipc2.m2(10));
18    }
19 }
```

Now write a Java Class named “**PA**” in the **PA.java** file, “**PB**” in the **PB.java** file, and, Interface named “**PC**” in the **PC.java** file according to their required packages, so that if we run the **main** method in **AQ2** class it gives the following output:

Output:

```
Start
ABC==ABC
10
15
25
Welcome Java
2
6
6
XYZXYZ
25
50
```

Consider the following class **AQ3** and the **main** method inside it. The following codes are provided to you in the **AQ3.java** file. You cannot change any code in this file.

```
1 //AQ3.java
2 public class AQ3 {
3     public static void main(String[] args) {
4         for (int n=0;n<6;n++) {
5             System.out.println("Exceptions Case: " + n);
6             try {
7                 if (n == 0) { System.out.println("Starting: "+n);
8                     Exc.a(); System.out.println("Ending: "+n); }
9                 if (n == 1) { System.out.println("Starting: "+n);
10                    Exc.b(); System.out.println("Ending: "+n); }
11                try { if (n == 2) { System.out.println("Starting: "+n);
12                    Exc.c(); System.out.println("Ending: "+n); }
13                    else if (n == 3) { System.out.println("Starting: "+n);
14                        Exc.d(); System.out.println("Ending: "+n); }
15                    else { System.out.println("Starting: "+n);
16                        throw new NewExc(n); }
17                } catch (ArrayIndexOutOfBoundsException ex) {
18                    System.out.println("Exception is "+ex);
19                    if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
20                }
21            } catch (ArithmeticException | NullPointerException ex) {
22                System.out.println("Exception = "+ex);
23                if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
24            } catch (NewExc ex) {
25                System.out.println(ex);
26                if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
27            } catch (Exception ex) {
28                System.out.println("Any Exception");
29                if(ex.getCause()!=null) System.out.println("Cause: "+ ex.getCause());
30            }
31        }
32        System.out.println("Exam Ended");
33    }
34 }
```

Now write a Java Class named “**Exc**” in the **Exc.java** file, and, “**NewExc**” in the **NewExc.java** file so that if we run the **main** method in **AQ3** class it gives the following output:

Output:

```
Exceptions Case: 0
Starting: 0
Exception = java.lang.ArithmeticException
Exceptions Case: 1
Starting: 1
Exception = java.lang.NullPointerException
Cause: java.lang.ArithmeticException
Exceptions Case: 2
Starting: 2
Exception is java.lang.ArrayIndexOutOfBoundsException: Index Out!
Exceptions Case: 3
Starting: 3
New Exception: New Exception: 0
Exceptions Case: 4
Starting: 4
New Exception: New Exception: 4
Exceptions Case: 5
Starting: 5
New Exception: New Exception: 5
Exam Ended
```